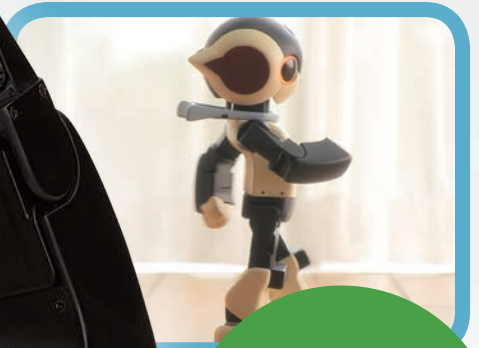
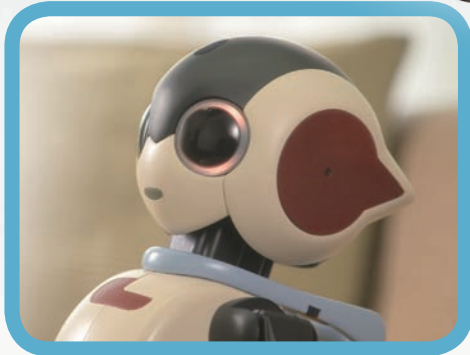
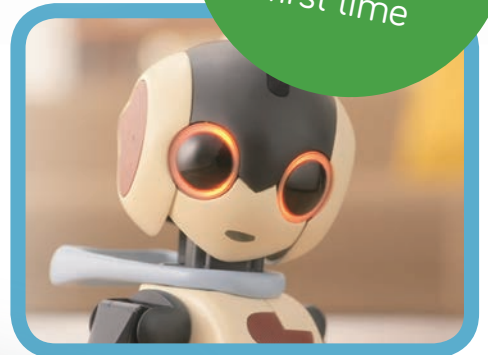


Build your own Robi

Pack **14**



Stages 51-54:
Test Robi's legs
together for the
first time



Build your own **Robi**

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Assembly Guide **p270**

Stage 51: Program and fit Robi's left hip servo **p270**

Stage 52: Join Robi's legs for the first time **p278**

Stage 53: Assemble Robi's neck base **p288**

Stage 54: Prepare Robi's neck servo **p293**

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PLAY. ITEMS MAY VARY FROM THOSE SHOWN. THERE MAY BE
COLOUR VARIATIONS OWING TO THE MATERIALS USED.

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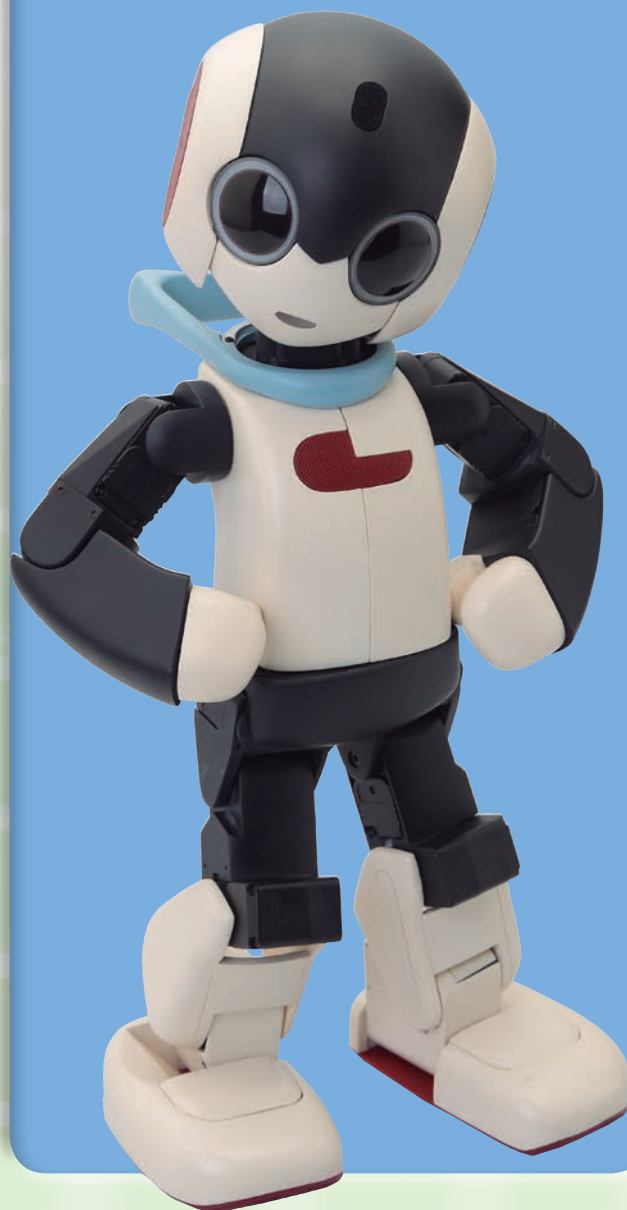
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STAGE 51: PROGRAM AND FIT ROBI'S LEFT HIP SERVO

Set the ID number of Robi's left hip servo, then fit and test it as part of the left leg assembly built over the previous stages.

This stage starts with the familiar task of wiring a servo and setting its ID number, in this case ID=7, Robi's left hip servo. Once that is complete, you will go on to fit this into the existing left leg assembly, which involves some

more fiddly work with servo cables, so pay especially close attention to the instructions. Finally, you will test the left leg, using the CPU board as you have been doing throughout the previous few packs.



1



YOUR PARTS

1 Servo motor (ID=7)

PARTS TO BE ASSEMBLED

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



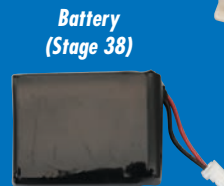
Left outer hip frame (Stage 50)



Neck stand (Stage 8)



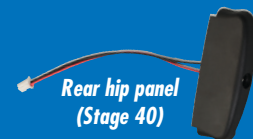
Left leg assembly (Stage 49)



Battery (Stage 38)



CPU board with test micro SD card inserted (Stage 44)



Rear hip panel (Stage 40)



Servo cable (Stage 50)

THE SERVO AND CABLE

TIP!

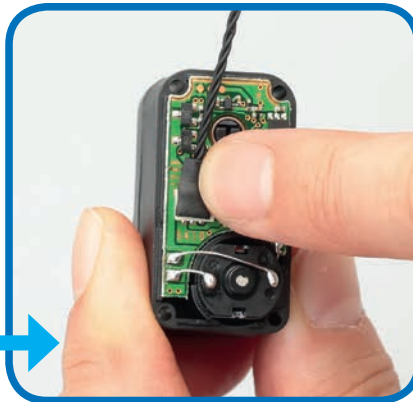


As ever, it is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Make sure to hold the open servo by its sides, and if you put it down on your work surface during the assembly, do this with the open side facing upwards.



1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.

2 Gently lift the cover away, taking care not to force it or damage the circuit board underneath.



TIP!



This servo cable had protective pads fitted to its connectors in the previous stage, but if you find it difficult to attach with a pad in place, it is possible to fit the pad afterwards. If you choose to do the assembly this way, be especially careful not to press down on the circled projection, next to the cable, as you do so.

3 Hold the open servo with the circuit board facing upwards. Fetch the servo cable fitted with protective pads in the previous stage, and press one of its connectors into the socket circled in red, as you have done before. It does not matter which end of the servo cable you use.

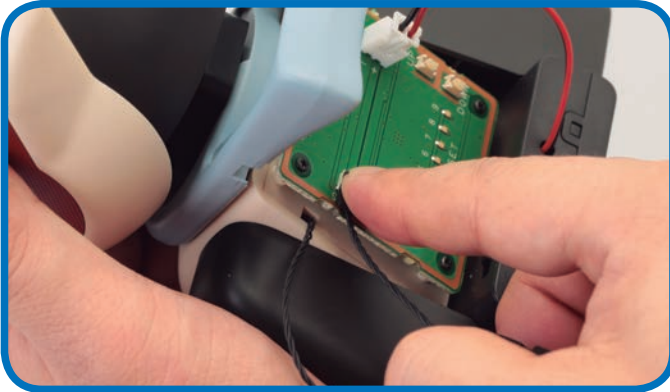


4 Pass the free end of the servo cable through the circular hole in the removable cover, as shown.

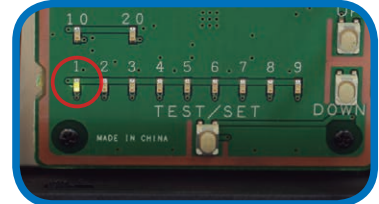
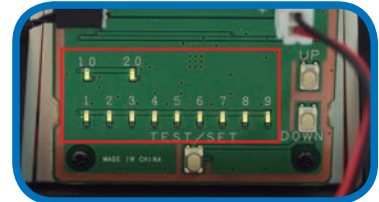
5 Press the cover back into place on the servo's main housing.

6 Fasten the cover back into place with all four screws, but only lightly, as you will be removing them again at a later stage.

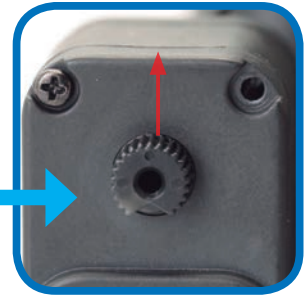
TESTING THE SERVO



- 7** Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board, and attach the servo cable connected to the servo in Step 3.

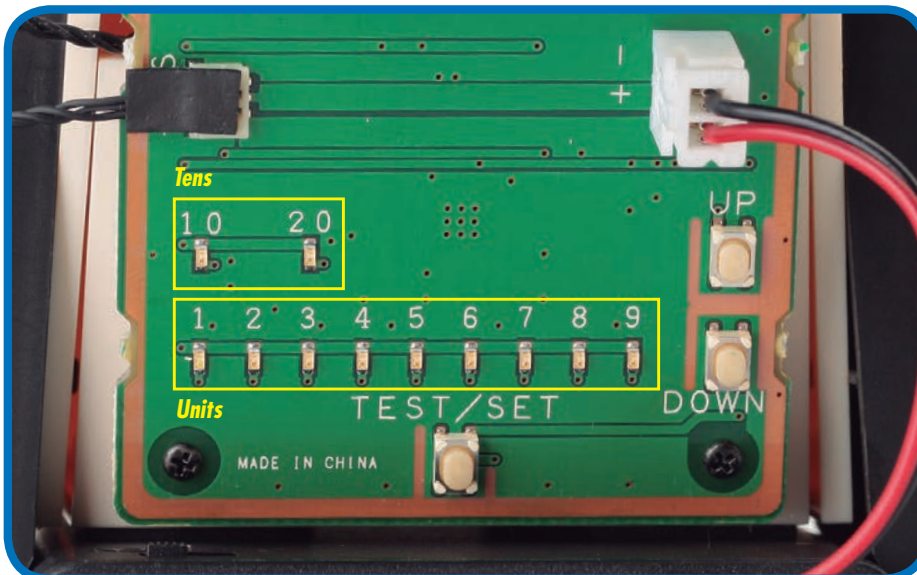


- 8** With the servo connected, switch the power to ON. After the LEDs blink twice, only the 1 LED should light up.

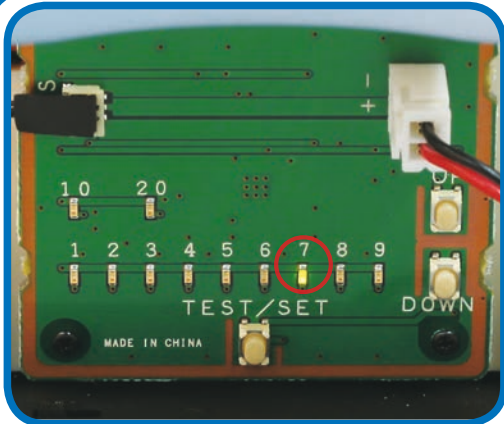


- 9** To test the servo, press the TEST/SET switch: the servo's shaft should rotate 45° to the left, stop, then move back across to 45° to the right. The next step is to set the servo's ID number, so keep the power on. If the test does not run, or if the LEDs flash rapidly, refer to the HELP box in Stage 8.

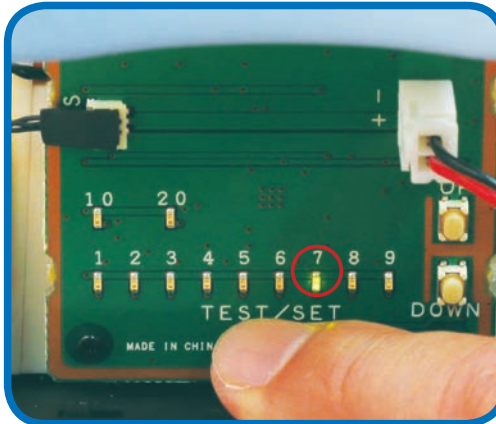
SETTING THE ID NUMBER



- 10** Next, you will set the ID number of the servo using the test board, as you have done before. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 7, as you will do in this stage, you press UP 6 times. This will light up the 7 LED alone. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, you set it by pressing and holding the TEST/SET switch.



- 11** To set the ID number of Robi's left hip servo to 7, press the UP button 6 times. Make sure that only the 7 LED is lit, using the UP and DOWN buttons to correct the number if necessary.

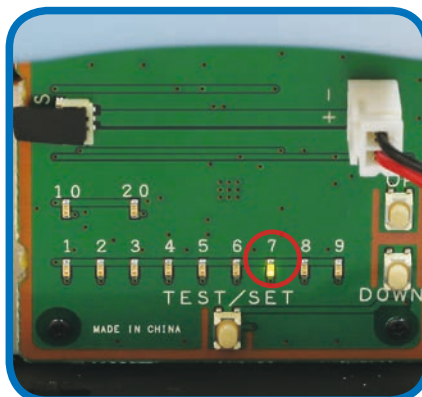
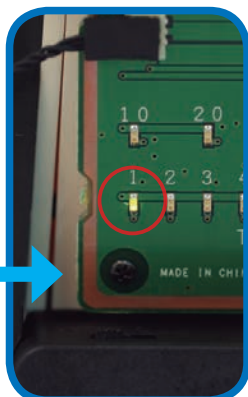
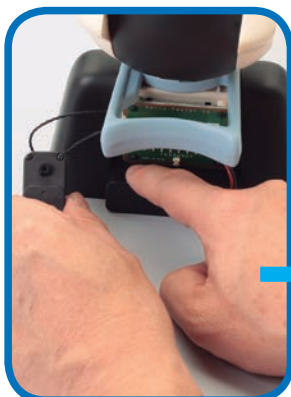


- 12** Confirm the setting by pressing and holding the TEST/SET button for a few seconds. The 7 LED should flash a few times, then remain lit.



ID=7

CHECKING THE ID NUMBER



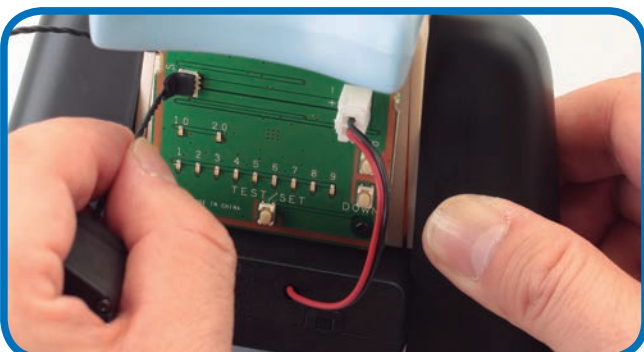
HELP!



It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake, or set the wrong ID number, you can reset it by going through the setting procedure again.

- 13** It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, then switch the power OFF, then back ON. Only the 1 LED should light up.

- 14** To check the ID number, press TEST/SET briefly. Now the 7 LED should light up and the servo shaft should repeat the rotations seen in the test. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.

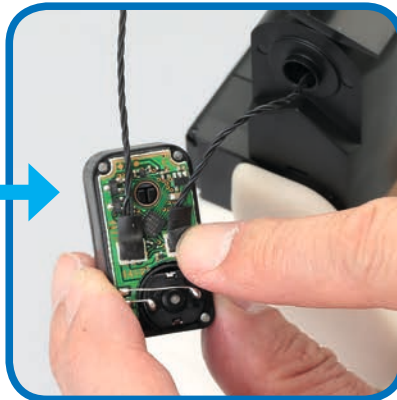


- 15** Once all is well, switch the power pack OFF, then carefully unplug the servo from the test board. It will retain its setting, ready to fit into Robi's leg.



- 16** Remove the screws fitted in Step 6 and take the cover off. Keep the screws, but you will not be using the cover again.

FITTING THE LEFT HIP SERVO



17 Line up the hip servo to the left leg, and fit the servo cable leading from the thigh servo into its free socket.



18 Making sure that both connectors are sound, line up the servo to the hip frame, as shown.



19 Press the servo into the frame. Be careful not to pinch or trap either cable as you do this.



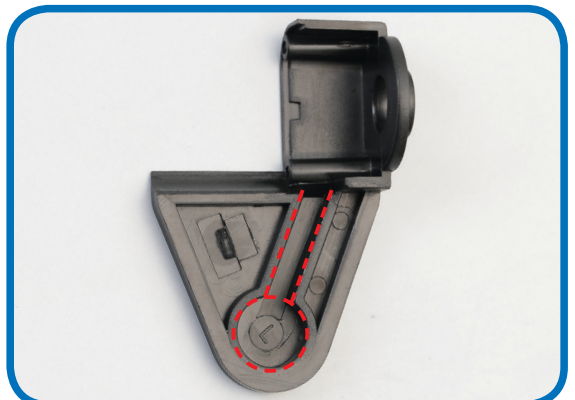
20 Use two of the screws removed in Step 16 to secure the servo via the two circled holes.



ADDING THE OUTER HIP FRAME



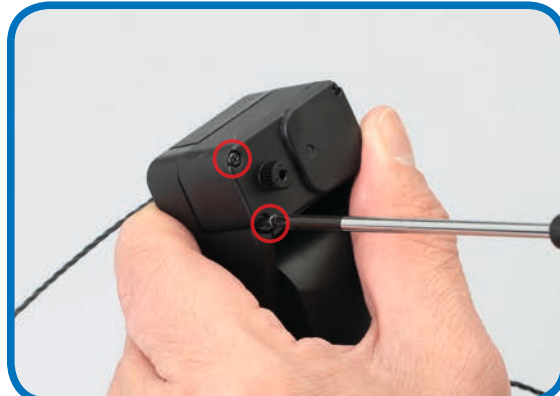
21 Orient the left outer hip frame as shown above. Then take the servo cable leading from the hip servo that you just fitted, and pass it through the circled hole in the inside of the part.



22 Identify the recessed area of the outer hip frame highlighted above. The servo cable will sit in this once fitted.



23 Seat the cable linking the thigh and hip servos into the recess highlighted in the previous step, then close the frame over the servo.

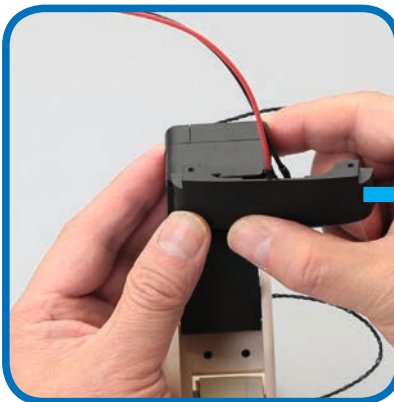


24 Use the other two screws removed in Step 16 to secure the frame.

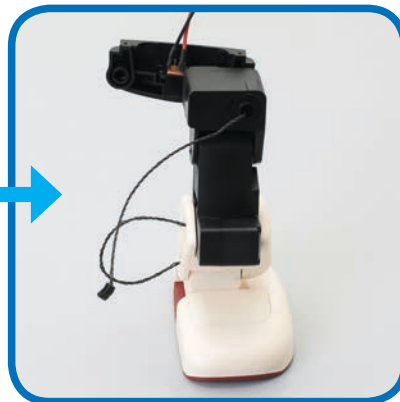
TESTING THE LEFT LEG AND HIP



25 Fetch the rear hip panel (with the power distribution board and cord) and line up its circled D-cut socket to the backward-facing hip servo's shaft. Remember, the D-cut parts will fit neatly and easily when set correctly.



26 When the parts are set as they should be (see the Close-Up boxes throughout the series for info on handling D-cuts safely), press the hip panel onto the shaft, then turn the leg around to face you.



27 Now fetch the CPU board with the test micro SD card inserted, and insert the hip servo's cable into the left leg socket on the board, the one that is second closest to the micro SD card slot.



28 Connect the battery to initiate the test – bear in mind that **the test will begin as soon as the battery is connected.**

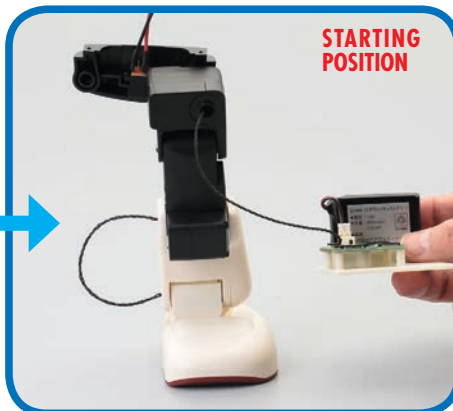
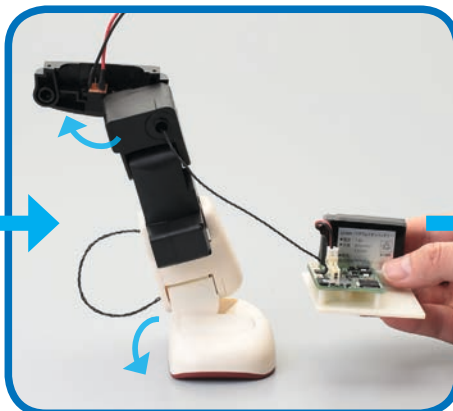
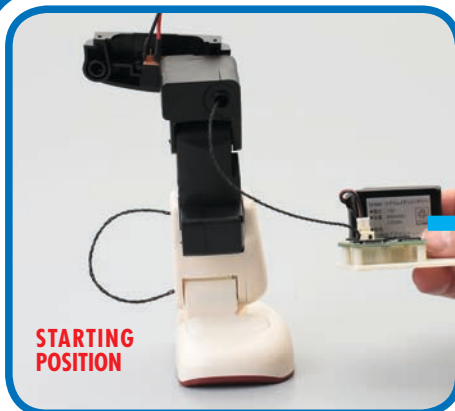
ATTENTION!

Robi's left leg will start moving as soon as you begin the test, so make sure you perform this test in a safe place where the leg will not knock anything over.

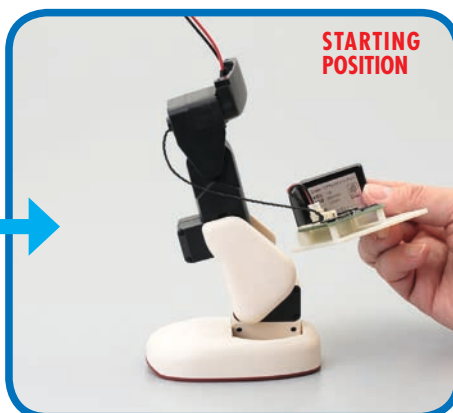
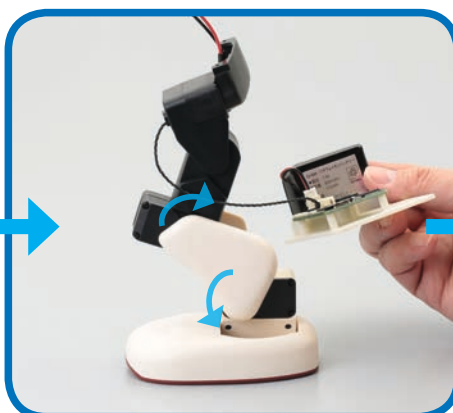
WARNING!

When you connect the battery to the CPU board, disconnect it immediately if you see any smoke or vapour rising from the board!





- 29** When the test begins, the leg will move to its starting, or neutral, position. Then, the foot and hip servos – as the two that rotate from side to side – will move together, with the hip servo fitted in this stage counteracting the inward tilt of the foot servo to keep the hip panel level. The servos will then return to their starting positions once more.



- 30** Next, the ankle, knee and thigh servos will begin their movement test by adjusting themselves to make Robi's left leg 'squat' down for a moment. They will then return to their starting positions. The test will continue to run until the battery is disconnected, so observe a few runs of it, then remove the battery to end the test. When you are happy that all is as it should be, remove the rear hip panel and store it safely until it is needed again.

If the test does not run as described, read the troubleshooting section in this stage to see how to address the issue.

As with other stages in which you connected two or more servos, it is vital that you do not attach the servos linked in this stage to the neck stand/test board! Doing so will wipe their ID numbers, meaning you will have to go back and open each servo to reset it.



Assembled left leg

Robi's left leg is now complete, ready to be joined to his right.



You have now completed both of Robi's arms and legs!

Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

TROUBLESHOOTING

In this section, we look again at what to do if the movement test carried out in this stage does not go as described.

Along with the Phillips screwdriver supplied with Stage 2, you will need the following:



Servo removal tool (Stage 41)

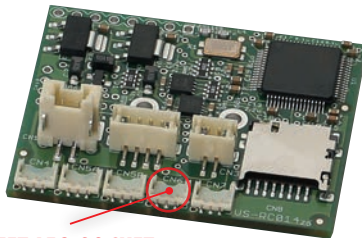


Neck stand (Stage 8)

IF NONE OF THE SERVOS MOVE

Before dismantling the leg to check the individual servos, check the other electrical components used in the test. Then, if that doesn't solve the problem, here's how to dismantle the subassembly safely and efficiently.

CHECK THAT THE LEG IS CONNECTED TO THE CORRECT SOCKET ON THE CPU



LEFT LEG SOCKET

The socket second closest to the micro SD card slot on the CPU board will hold the left leg's servo cable.

CHECK THAT THE SERVO CABLE IS SECURELY CONNECTED TO THE CPU



Ensure that the left leg's servo cable connector is sitting securely within its socket. Remove and replace it carefully if need be.

CHECK THAT THE BATTERY IS CHARGED



The test will only run if sufficient power is received from the battery, so make sure the battery is fully charged by following the steps explained in Stage 41.

IF NO SERVOS MOVE

If none of the servos move when you perform the test, it is likely that there is a connection fault with the hip servo, and resetting its servo cable connections may well resolve this. If not, you will need to dismantle the leg and check the other connections and servo numbers.



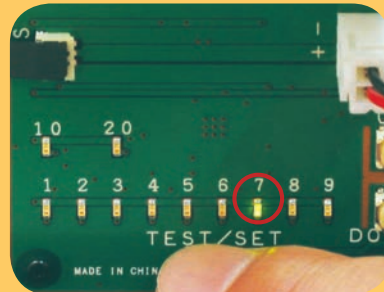
DISASSEMBLY

Follow the assembly of Steps 17-24 of this stage in reverse, and check the hip servo's connections carefully. If this does not solve the problem, you may need to continue to dismantle the thigh, knee, ankle and foot

and test/reset their faulty servo(s) as necessary. Stages 45, 46, 48 and 49 deal with this in detail.

IF THE HIP SERVO DOES NOT MOVE

By now, you will have tested all of the left leg's servos, so these should be sound (see Stage 49 if this is not the case). If these servos move, but the hip servo does not, follow Steps 7-24 of this stage in reverse order and check/reset the hip servo's ID number.



DISASSEMBLY

Follow the assembly of Steps 7-24 of this stage in reverse order and, if all the connections are sound, test/reset the hip servo. Its ID number should be 7, and more information on the

process of resetting this servo's ID number is given earlier in this stage. If any other of the leg's servos do not move, see the Disassembly box to the left.

STAGE 52: JOIN ROBI'S LEGS FOR THE FIRST TIME

In this stage, you will join Robi's legs at the hip, and then perform a combined movement test using the CPU board.

On completion, the servo cables leading to each of Robi's limbs and his head will fit into their respective sockets on the CPU board – much as they have done during the tests – and each body part needs to be wired carefully and efficiently. In this stage, you will join

Robi's legs, feeding their cables, along with the power cord set into the rear hip panel, through the hip assembly and into where Robi's CPU will be fitted on completion. You will then test both legs together. You have also received a ball with which Robi will play soccer!



PARTS TO BE ASSEMBLED

YOUR PARTS

- 1 Ball
- 2 Hip base
- 3 Front hip panel
- 4 M2 x 6mm countersunk screws x 4 (one is a spare)

USEFUL TOOLS

Servo removal tool
Tweezers
Tape

SAVED PARTS

You will be using parts from previous stages, so be sure to have these to hand before beginning.



*Battery
(Stage 38)*



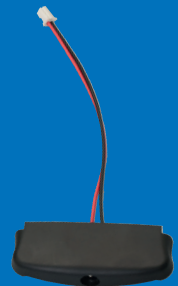
*CPU board with test
micro SD card inserted
(Stage 44)*



*Right leg
(Stage 32)*



*Left leg
(Stage 49)*

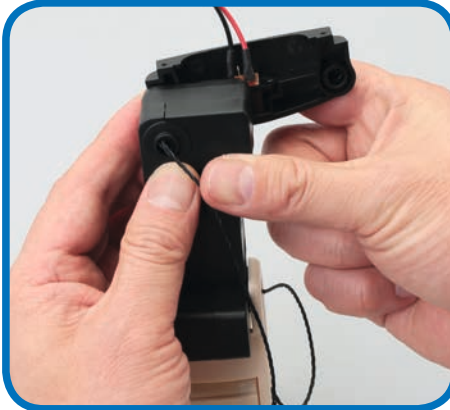


*Rear hip panel
(Stage 40)*

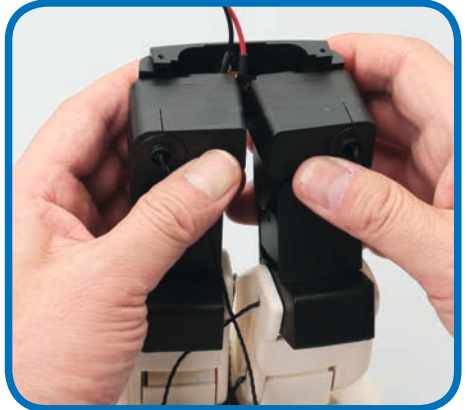
JOIN THE LEGS



1 Line up the rear hip panel to the backward-facing shaft of the right hip servo.

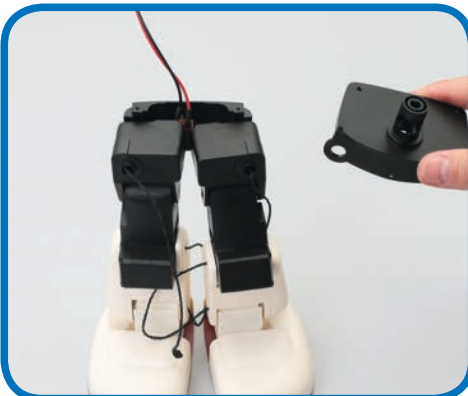


2 Making sure the D-cut shaft and socket are aligned correctly, press the parts together.

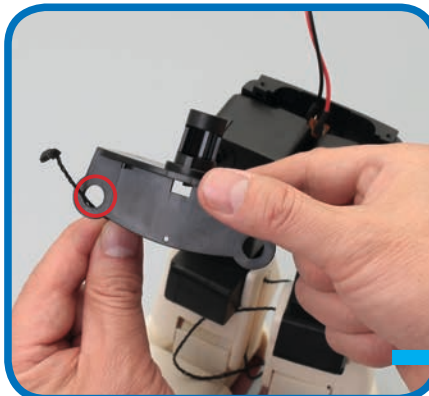


3 Repeat Steps 1-2 for the left leg, again making sure the D-cut sections are set correctly.

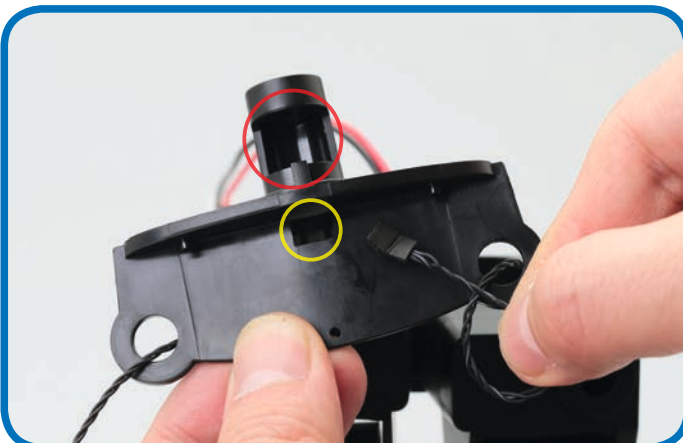
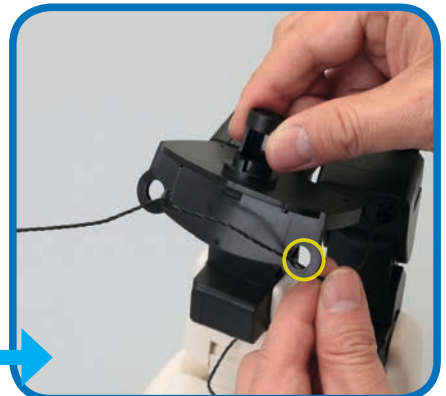
WIRING THE HIP BASE (SERVO CABLES)



4 Set the legs out level on your work surface and then line up the hip base to them, as shown.



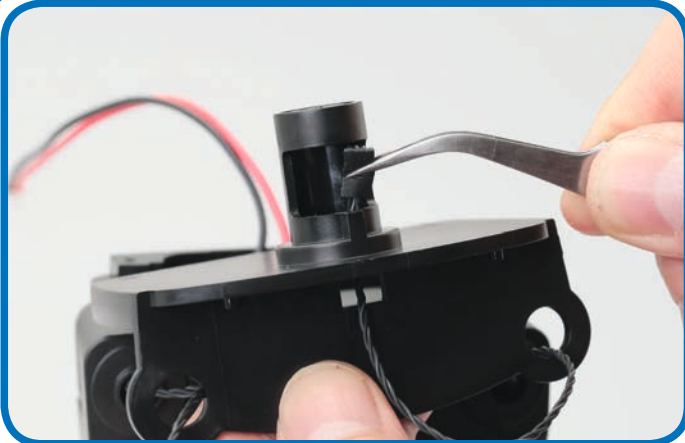
5 Feed the servo cable leading from the right leg through the red-circled hole, and the one leading from the left leg through the yellow-circled one.



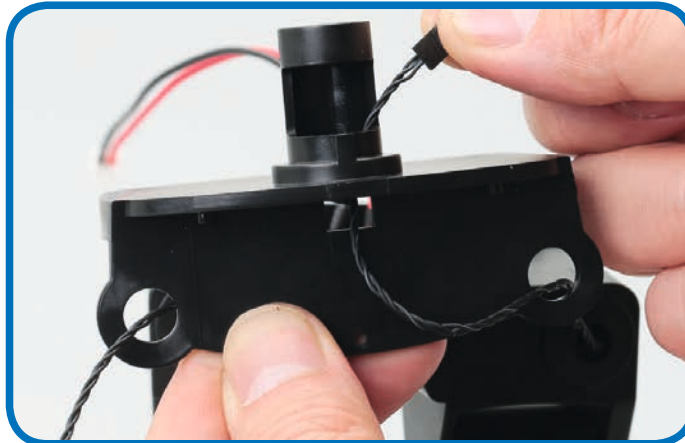
6 Now run both cables back into the hip base through the yellow-circled hole, then out through the larger, red-circled one. Begin with the left leg's cable.



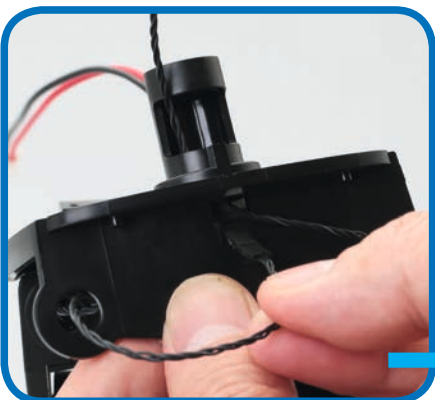
7 Feed the cable in by hand.



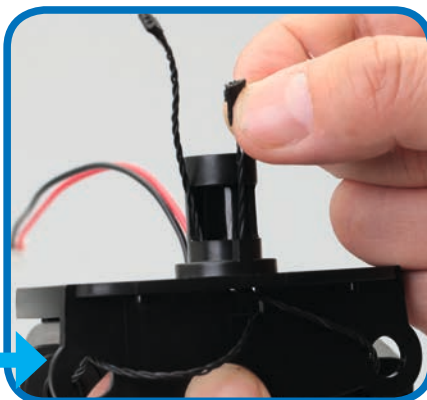
8 Using tweezers will make it easier to run the cable out of the larger hole, as shown above. **Be very careful not to pinch the servo cable connector!** Hold it by its protective pad or lightly by the cable.



9 Now take the cable in your fingertips and work it through to minimise slack and ensure that it does not fall back into the hole.



10 Making sure the left cable does not fall out, repeat the process for the right leg's servo cable.

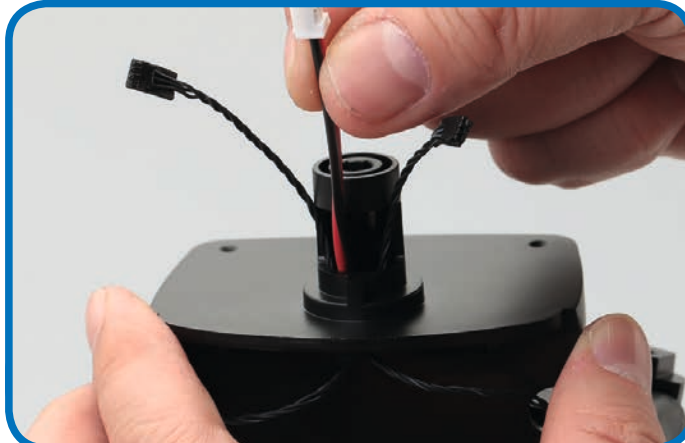


11 Your assembly should look like this.

WIRING THE HIP BASE (POWER CORD)

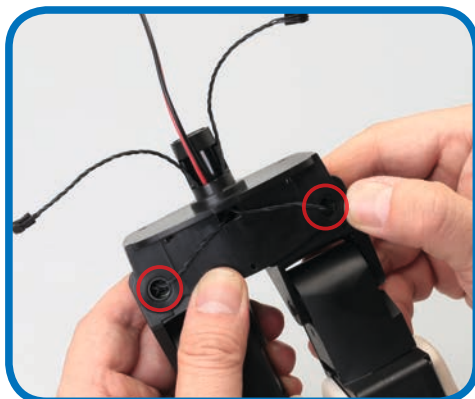


12 Making sure the servo cables do not fall out, tilt the hip base so that you can access its underside, as shown. Then line up the rear hip panel and feed its power cord through the arrowed hole.



13 Turn the hip base back over, then gently pull the white connector of the power cord upwards out of the same larger rectangular hole as the servo cables.

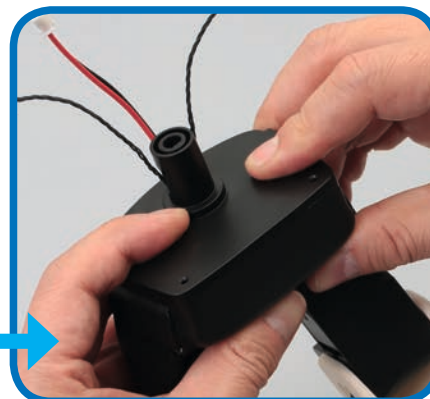
SECURING THE HIP BASE



14 Join the hip base to the legs, with the two circled holes neatly aligned.



15 Pull the servo cables and power cord gently upwards enough to ensure they remain in place, then press the hip base firmly onto the legs.



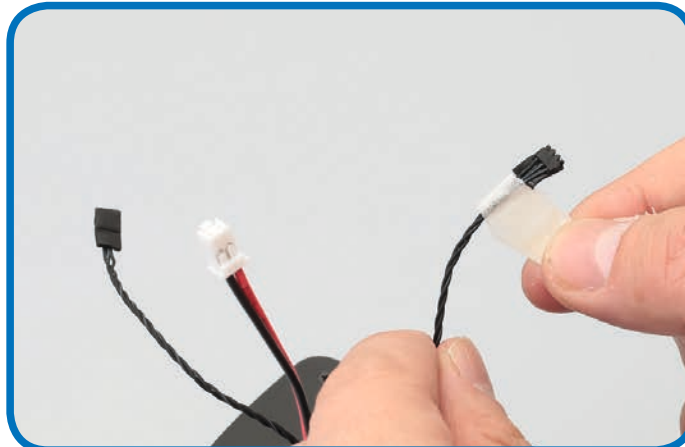
16 Make sure the circled holes are clear, then insert and tighten two M2 x 6mm countersunk screws to secure the base.



TAPING THE RIGHT CABLE

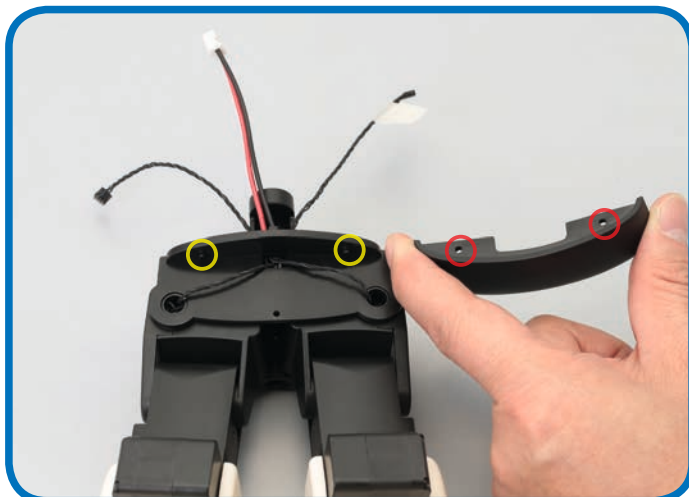


17 Locate the servo cable leading from the right hip.



18 Use a short length of tape to mark this. Wrap the tape over the cable itself, not the pad or connector.

SEALING THE HIP PANELS



19 Line up the front hip panel to the assembly. The circled pins and holes will meet when fitted.



20 Join the parts, taking care not to interfere with any of the servo cables as you do so.



21 Look into the hip assembly from the underside and use a fingertip to push the red and black wires of the power cord into the central groove. This will prevent them from getting in the way of the two hip servos.

TIP!

If you find the power cord's connector is preventing the hip frames closing neatly, you can bend the cord inwards a little. Only do this if absolutely necessary, and if you do, be extremely careful not to break the prongs.



22 Position the hip and leg assembly with the legs separated as far as they will go, as shown. Then locate the circled countersunk hole in the front hip panel, and insert and tighten another of the M2 x 6mm countersunk screws into it to secure the parts.

TESTING THE LEGS AND HIPS

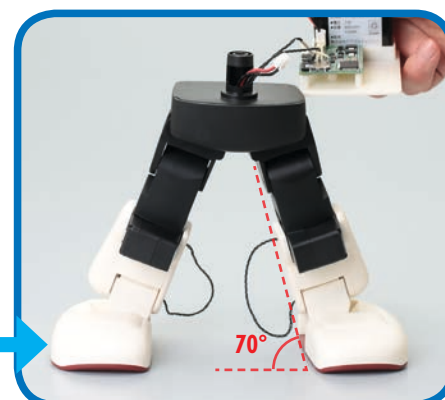
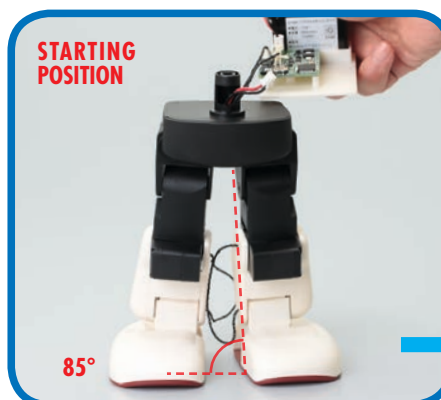
ATTENTION!

Robi's legs will start moving as soon as you begin the test, so make sure you perform this test in a safe place where they will not knock anything over.

WARNING!

When you connect the battery to the CPU board, disconnect it immediately if you see any smoke or vapour rising from the board!

- 23** Insert the connector of the right leg servo cable (marked by the tape) into its socket on the CPU board, the one closest to the micro SD card. Then insert the left leg's connector into the socket next to it.



- 24** Place the legs on your work surface with plenty of space around them, then connect the battery pack to the CPU board. **The legs will start the test as soon as the power is connected, so be ready.**

- 25** The legs will begin by resetting themselves to the starting, or neutral, position. It's a good idea to make a note of what the angle should be – about 85° to your work surface, almost perpendicular – in case you find you need to adjust them later (see the troubleshooting guide of this stage if this is the case). The legs will then simultaneously slide outwards, to an angle of roughly 70°, and then return again to the starting position.



Assembled legs

Robi's legs are now joined at the hip and ready for the next phases of the build.

- 26** Now watch from the side as the legs perform a combined squat. The test will repeat until you disconnect the battery, so watch a few times to ensure the legs are moving and returning to the starting position correctly, then disconnect the battery. If the test is not running as it should, or the legs are not returning to their neutral positions at the start, see the troubleshooting guide for how to remedy this.



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

TROUBLESHOOTING

In this section, we look again at what to do if the movement test carried out in this stage does not go as it should. As with the previous tests you have performed, there is a likelihood that checking/reattaching the servo's connections will resolve the issue quickly and easily. If necessary, though, you may need to dismantle the problem leg and test/reset its servo ID numbers. Another problem we will look at here is what to do if the legs or feet are not setting to their starting or neutral positions as they should, in which case it is probable that the D-cut parts joining the components need attention.

ATTENTION!

The photos in the following steps show the left leg, but the instructions, suitably adapted, also apply to the right leg. So if it is the right leg that is faulty, just follow the same basic procedure.

Incorrectly set neutral positions

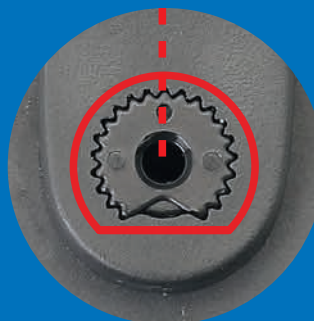
When the test begins, the legs should reset to their neutral positions, which are marked in the step photos for each test section. Observe the legs' neutral positions closely from the front and side to assess if they are as they should be. Disconnecting the battery as soon as the legs move into the neutral position will freeze the test, allowing you to lift them up by the hip base to inspect them more easily (see below).

If any of the joints does not resemble those shown in the photos, it is probable that it was not set correctly when fitted. To remedy this, you need to check/reset the D-cut joint in question (see box, right). Be careful not to repeat the mistake when you reset the joints.

THE D-CUT



The D-cut components are the key to holding Robi's joints in the correct neutral positions, so they must be set correctly from the start for Robi to function properly! If, when you perform the tests, one or other joint is askew, you will need to dismantle it and reset its joint to the correct position.



The D-cut hole holds the servo shaft straight.



If not set correctly, the joint will be off-centre, and the plastic can be damaged.

Disconnect the power as soon as the legs reach their neutral positions, to give yourself more time to inspect them.



IF THE FOOT IS NOT FLAT (FROM SIDE TO SIDE)

If you look at Robi's legs from the front when they are in their neutral positions and see that one or other foot is not sitting flat on your work surface, it is the ankle that needs attention.



1 Remove the three screws holding the sole of the problem foot. **Keep these screws safely.**



2 Now remove the foot cover.



3 Next, remove the two screws securing the ankle joint to the sole. **Again, keep these, making a note of which ones they are.**



4 Now remove the ankle frame from the rest of the foot, and check the D-cut hole at the front of the part, which joins it to the foot servo's shaft. Rejoin the two, making sure that the joint is neat and clean, then repeat these steps in reverse to reassemble the foot.



For further instructions on the process of reassembling and/or resetting the ID numbers of Robi's feet and ankles, see Stages 23-27 (right) or Stages 35, 36, 37 and 45 (left).

IF THE FOOT IS NOT FLAT (FROM FRONT TO BACK)

If, when you look at Robi's legs from the side when they are in their neutral positions, you see that one or other of the feet is tilting forward or back, it is the lower leg that needs attention.



5 Remove the two screws holding the inner and outer lower leg frames together. **Keep the screws.**



6 Remove the inner leg frame, making sure not to tug on the servo cable as you do so.



7 You will now be able to separate the outer lower leg frame from the ankle servo's shaft.



8 Check and refit the lower leg frame onto the servo shaft, making absolutely sure it is on correctly. Repeat these steps in reverse to reassemble the ankle and lower leg.

For further instructions on the process of reassembling and/or resetting the ID numbers of Robi's lower legs and ankles, see Stages 26-28 (right) or Stages 44-46 (left).

IF THE LOWER LEG IS AT THE WRONG ANGLE (FROM FRONT TO BACK)

If it is one of Robi's lower legs that is askew when viewed from the side, you will need to dismantle and reset that leg's knee joint.



9 Repeat Steps 5 and 6, but this time it is the D-cut joint between the lower leg frame and the knee servo, rather than the ankle servo, that needs to be reset. Do this, then reassemble the leg.

For further instructions on the process of reassembling and/or resetting the ID numbers of Robi's knees, see Stages 29-30 (right) or Stages 47-48 (left).

IF THE LEG IS AT THE WRONG ANGLE FROM THE THIGH DOWN (FROM FRONT TO BACK)

If one of Robi's legs is tilting too far forward or back from the thigh, it is the joint at the hip servo that needs attention.



10 Check the D-cut joint between the thigh servo and the inner hip frame. No dismantling is required for this, but if you find that the joint is not fitted as it should be, you will have to take it apart and reset it, as described in the following steps.



11 If you need to reset the joint between the thigh servo and the hip frame, remove the two circled screws holding the inner hip frame to the hip servo. (Removing all four of the hip servo's screws may make it easier if you find it difficult to lift the frame away after removing only two).

12 You will now be able to remove and refit the inner hip frame to the thigh servo's servo shaft.

For further instructions on the process of reassembling and/or resetting the ID numbers of Robi's thighs, see Stages 31, 32 and 50 (right) or Stages 49-51 (left).

IF THE LEG IS AT THE WRONG ANGLE FROM THE THIGH DOWN (FROM SIDE TO SIDE)

If one or other of Robi's legs is sticking out to the side when viewed from the front, you will need to dismantle the hip and refit the D-cut joint between the hip servo and rear hip panel.



13 Begin by removing the two screws holding the hip base to the rear hip panel.



14 Lift the hip base away from the rest of the hip assembly.



15 You can now remove the rear hip panel, freeing the hip servo so that you can check and reset the D-cut joint.

For further instructions on the process of reassembling and/or resetting the ID numbers of Robi's hips, see Stages 33 and 51 (right) or Stage 52 (left).

STAGE 53: ASSEMBLE ROBI'S NECK BASE

In this session, you will begin the process of building up Robi's head and neck, by removing Robi's 'scarf' from the neck stand and fitting the new neck base supplied with this stage.

In this session, you will continue working on Robi's head and neck, which you last did way back in Packs 1 and 2 when you assembled his eyes and built up his head. This time, you will dismantle the neck stand to

separate the scarf from the existing neck base, before securing it to the new one supplied with this stage. You will replace the servo and neck stand base, as you will be using its test board again in the coming stages.

PARTS TO BE ASSEMBLED



1



2



3



YOUR PARTS

1. Neck base
2. Washers (one is a spare)
3. Servo cable (135mm)

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



*Protective pads
(Stage 3)*



*Neck stand
(Stage 8)*

DISMANTLE THE NECK STAND



1 Remove the servo cable from the test board.



2 Hold the neck stand and carefully lift the head from it.



3 Remove the four circled screws from the neck stand's base and keep these safely to one side; you will be using them again.



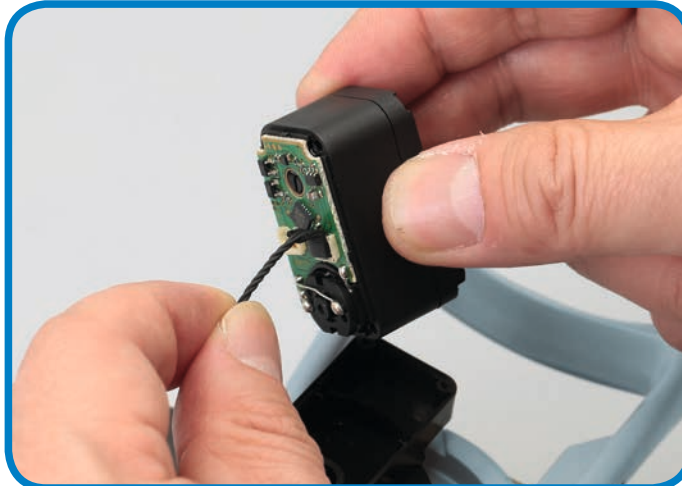
4 Lift the base and neck servo from Robi's scarf and neck stand. Hold the servo cable as shown to stop it slipping inside the stand, which will make the steps later in this session more difficult.



5 Remove the four circled screws, again keeping hold of them for later use.



- 6** Remove the servo from the neck base. Be sure to hold the open servo by its sides to avoid touching the exposed electronic components.



- 7** Carefully unplug the servo cable from the servo's socket, still making sure not to touch the electronic components.



- 8** Separate the neck base from the rest of the assembly.



- 9** Now lift the scarf from the neck stand. Again, hold the servo cable in place to stop it slipping inside the stand.

REASSEMBLE THE NECK STAND



- 10** Feed the servo cable – still running through the neck stand – back through the central hole in the neck base.



- 11** Carefully re-insert the servo cable's connector into the servo's socket.



- 12** Join the servo and neck base, with the shaft directly above the centre of the neck base, as shown.



- 13** Use the four screws that you removed in Step 5 to secure the parts.



- 14** Place the neck base and servo back on the neck stand and use the four screws that you removed in Step 3 to secure the parts.



- 15** Now replace the head, **making sure to fit it perfectly straight and to engage the D-cut servo shaft and its socket correctly.**

FITTING THE NEW NECK BASE



- 16** Line up the neck base supplied in this stage to the scarf, with the parts aligned as shown. Make a note of the groove around the raised central section (inset).

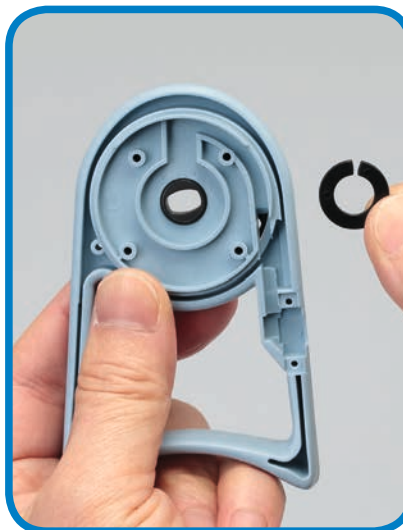


- 17** Fit the neck base into the matching recess in the scarf, as shown.

FITTING THE WASHERS

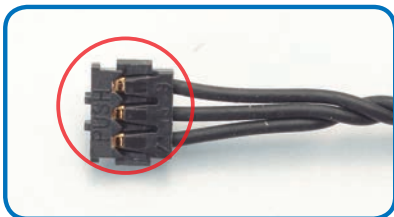
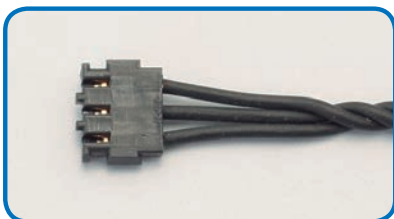


- 18** Inspect the washer closely – on one side there are three small circular moulding marks (the exact positions of these may differ from those shown here). It is this side that will touch the scarf when you fit the washer to it in the next step.



- 19** Position the scarf with its underside facing you, and keep the neck base in position by holding it from the back with your forefinger. Then fit the washer over the protruding central section of the neck base so that it snaps tightly into the groove around it (see inset, Step 16). Fitting the part of the washer with the gap in it first, then pressing home the rest of its circumference, will make this easier.

SEALING THE SERVO CABLE



- 20** As you have done previously, identify the side of this stage's servo cable connector that has PUSH marked on it (circled).
- 21** Apply a protective pad to the connector on this side. Repeat for the other end of the cable.

If you want to delay fitting the pads until the cable is used in the build, you may do so.

TIP!

Assembled neck base and scarf

The neck base that will hold Robi's neck servo on completion is now fitted to the scarf.



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

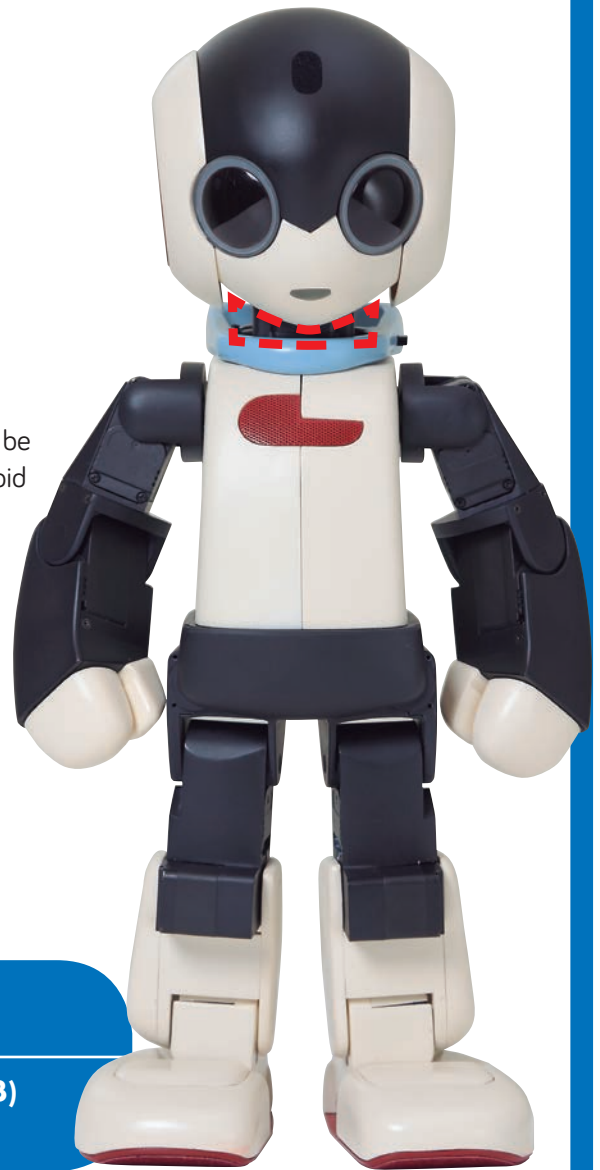
www.model-space.com

STAGE 54: PREPARE ROBI'S NECK SERVO

In this session, you will prepare the servo that will move Robi's head from side to side, servo ID=13.

This session sees you return to wiring and programming servos, in this case, the neck servo that will rotate Robi's head from side to side, servo ID=13. The steps are the same as for the

other servos you have prepared throughout the build, but, as ever, be sure to follow them closely to avoid having to go back and redo the steps at a later stage.



1



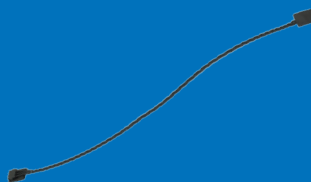
YOUR PARTS

1 Servo motor (ID=13)

**PARTS TO BE
ASSEMBLED**

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



Servo cable
(Stage 53)



Neck stand
(Stage 8)

THE SERVO AND CABLE

TIP!

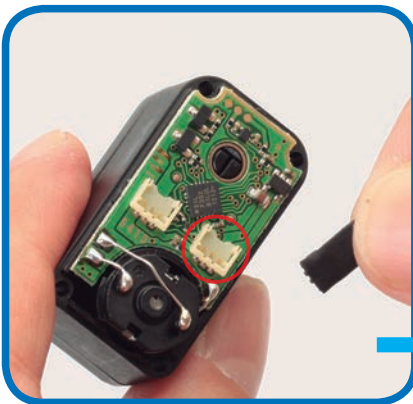


As ever, it is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Make sure you hold the open servo by its sides, and if you put it down on your work surface during the assembly, do so with the open side facing upwards.



1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.

2 Gently lift the cover away, taking care not to force it or damage the circuit board underneath.



TIP!



This servo cable had protective pads fitted to its connectors in the previous stage, but if you find it difficult to attach it with a pad in place, it is possible to apply the pad afterwards. If you choose to do the assembly this way, be especially careful not to press down on the circled projection, next to the cable, as you do so.

3 Hold the open servo with the circuit board facing upwards. Fetch the servo cable fitted with protective pads in the previous stage, and press one of its connectors into the socket circled in red, as you have done before. It does not matter which end of the servo cable you use.

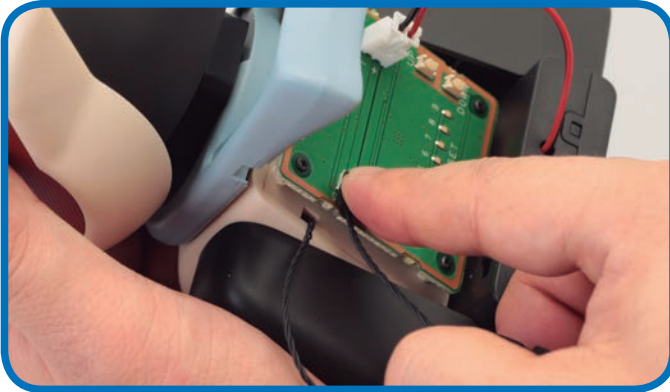


4 Pass the free end of the servo cable through the circular hole in the removable cover, as shown.

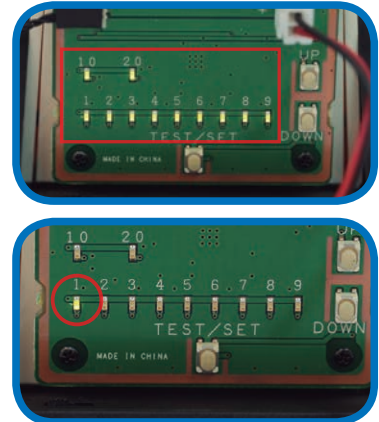
5 Press the cover back into place on the servo's main housing.

6 Fasten the cover back in place with all four screws, but tighten them only lightly as you will be removing them again later.

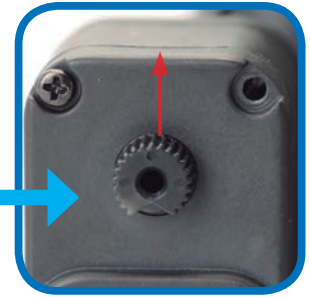
TESTING THE SERVO



- 7** Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board and attach the servo cable connected to the servo in Step 3.

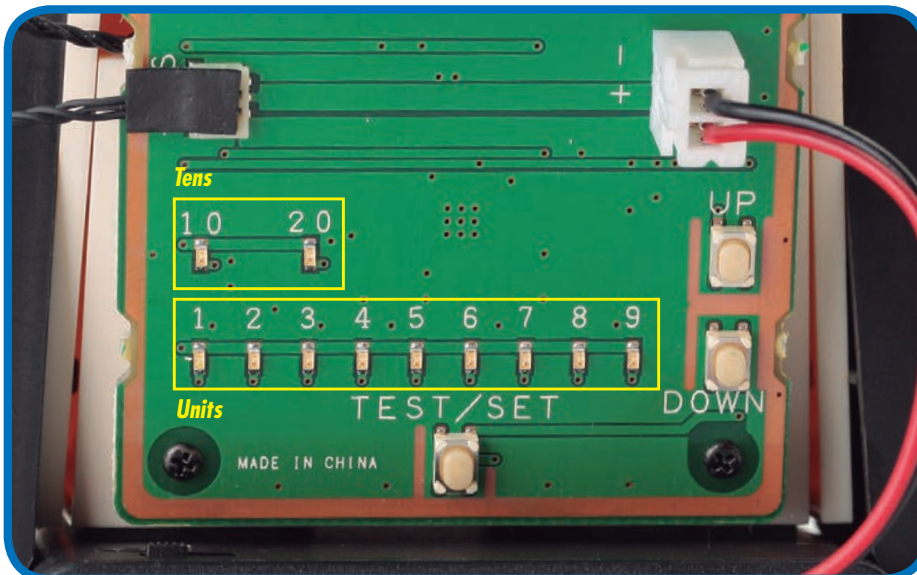


- 8** With the servo connected, switch the power to ON. After the LEDs blink twice, only the 1 LED should light up.

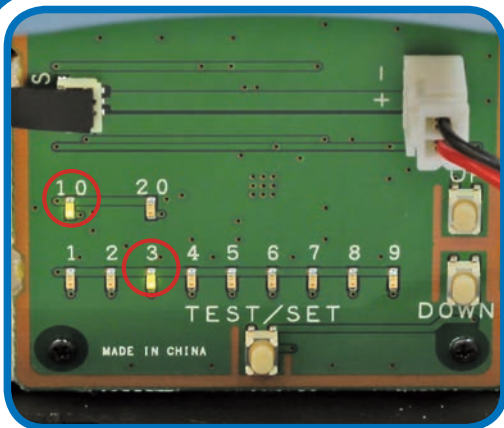


- 9** To test the servo, press the TEST/SET switch: the servo's shaft should rotate 45° to the left, stop, then move back across to 45° to the right. The next step is to set the servo's ID number, so keep the power on. If the test does not run, or if the LEDs flash rapidly, refer to the HELP box in Stage 8.

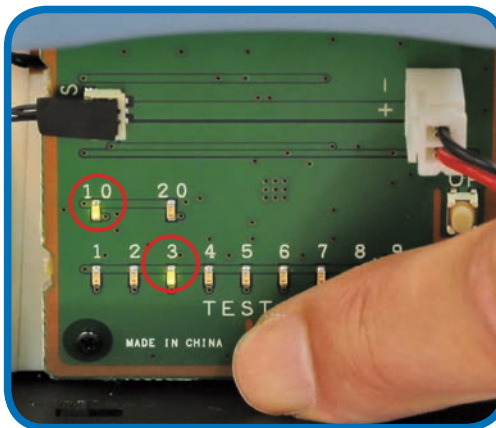
SETTING THE ID NUMBER



- 10** Next, you will set the ID number of the servo using the test board, as you have done before. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 13, as you will do in this stage, you press UP 12 times. This will light up the 10 and 3 LEDs. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, set it by pressing and holding the TEST/SET switch.



- 11** To set the ID number of Robi's neck servo to 13, press the UP button 12 times. Make sure that only the 10 and 3 LEDs are lit, using the UP and DOWN buttons to correct the number if you overshoot when setting.

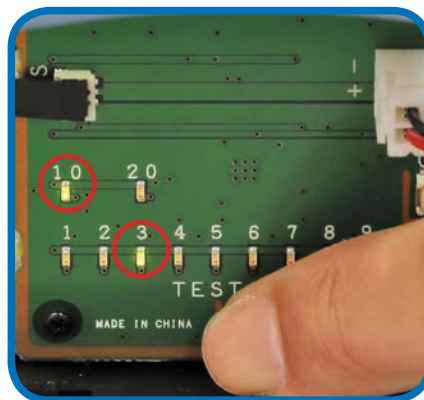
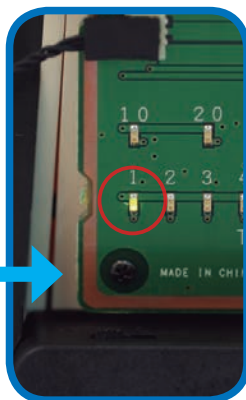
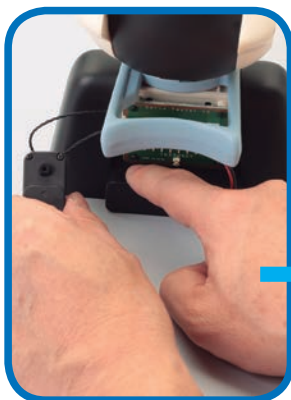


- 12** Confirm the setting by pressing and holding the TEST/SET button for a few seconds. The 10 and 3 LEDs should flash a few times, then remain lit.



ID=13

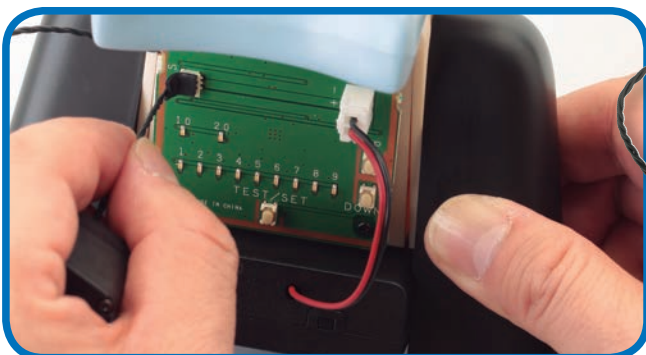
CHECKING THE ID NUMBER



It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake, or set the wrong ID number, you can reset it by going through the setting procedure again.

- 13** It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, then switch the power OFF, then back ON. Only the 1 LED should light up.

- 14** To check the ID number, press TEST/SET briefly. Now the 10 and 3 LEDs should light up and the servo shaft should repeat the rotations seen in the test. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.



- 15** Once all is well, switch the power pack OFF, then carefully unplug the servo from the test board. It will retain its setting, ready to fit into Robi's body.



Prepared servo

The neck servo is wired and its ID number (13) is set – store it away safely until it is needed. It is a good idea to write the ID number on it for ease of reference.



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

COMING IN PACK 15

Your next **FOUR** complete Stages, as Robi starts to come to life!



Stage 55
Assemble
Robi's speaker

PARTS PROVIDED
The speaker and
speaker covers.



Stage 56
Wire Robi's neck
base and scarf

PARTS PROVIDED
The cable for Robi's voice-recognition
card, a servo horn and cable.



Stage 57
Attach Robi's
ON/OFF switch

PARTS PROVIDED
Robi's switch and cable, a servo
cable, neck plate and screws.



Stage 58
Prepare Robi's
back servo

PARTS PROVIDED
The servo that will move
Robi's body, ID=12.